

Nishad Anil

📧 Senior AI/ML Engineer

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🌐 Portfolio

Professional Summary

- **Senior AI Engineer** with **4.5+ years** of experience designing and deploying production-scale Generative AI, Agentic AI, Computer Vision, NLP solutions across Fortune 500 industrial, renewable energy, banking, and healthcare domains.
- **Specialized** in Multi-Agent Systems, Advanced RAG Architectures, autonomous LLM applications, and cloud-native AI platforms, with proven experience delivering end-to-end AI products from research and model development to Kubernetes-based production deployment on AWS.
- **Hands-on expertise** in LangGraph, LangChain, OpenAI, MCP, LLM Finetuning, PEFT with LORA, Vector Databases, FastAPI, Kubernetes, AWS, YOLO, and LLMOps. Successfully improved model performance from **80% to 95% F1-score** and reduced inference latency by **40%** through optimization, automation, and scalable AI system design.
- **Experienced in building production-grade AI platforms**, integrating Agentic AI workflows, retrieval systems, and enterprise automation to improve decision-making, operational efficiency, and business outcomes.

Experience

Mphasis – Senior Software Developer

Jan 2026 – Present

- Developing and deploying a scalable report automation solution on AWS ECS for validating and comparing weekly operational reports across multiple business teams.
- Designed **Python and Pandas**-based data processing workflows integrated with ServiceNow ticket validation, enabling automated discrepancy detection, compliance reporting, and operational audit tracking.

Optimized Solutions Limited – Senior AI/ML Engineer

Apr 2025 – Jan 2026

- **Architected and delivered** a production-grade AI platform for a Fortune 500 industrial client, integrating Computer Vision, OCR, FastAPI, Pandas, and AWS cloud infrastructure to automate complex engineering document analysis.
- **Designed and implemented** an Agentic AI ecosystem utilizing LangGraph, MCP (Model Context Protocol) connectors, Advanced RAG pipelines, and SQL Agents to automate knowledge retrieval and engineering decision support across structured and unstructured enterprise data sources.
- **Developed an intelligent document-processing pipeline** using YOLOv11, AWS Textract, Pandas, and FastAPI to extract, validate, and structure data from complex engineering PDFs at scale.
- **Improved system performance by 40%** (from 8 minutes to 5 minutes per document) through model quantization, inference optimization, multiprocessing, and efficient data-processing workflows.
- **Improved object detection performance from 80% to 95% F1-score** through advanced augmentation techniques, dataset balancing strategies, and iterative model optimization.
- **Containerized, deployed, and scaled** AI workloads on Amazon EKS, collaborating with cross-functional teams to deliver secure, highly available enterprise systems supporting thousands of automated document-processing operations.

JSR Renewable Consultants – Senior Machine Learning Engineer

Jul 2023 – Apr 2025

- **Engineered an RAG-based knowledge assistant** for automated PDF error diagnosis, reducing manual troubleshooting effort and accelerating technical issue resolution across large document repositories.
- **Architected a Multimodal RAG pipeline** leveraging semantic chunking, hybrid search, Cross-Encoder reranking, table parsing, and image indexing to improve retrieval accuracy and minimize hallucinations.
- **Designed secure on-premise deployment architectures** enabling low-latency inference, air-gapped document processing, and compliance with enterprise data governance requirements.
- **Optimized custom YOLO models**, improving rare-class defect detection accuracy from **76% to 94%** and achieving **90%+ F1-scores** across industrial defect categories through advanced augmentation and class balancing techniques.
- **Developed OpenCV-based inspection pipelines** for automated measurement, annotation, and quality assurance reporting, enabling accurate pixel-to-millimeter dimension calculations for industrial applications.

Kshatra Infotech Pvt. Ltd. – Machine Learning Developer

Apr 2022 – May 2023

- **Developed and deployed** an end-to-end AI-powered document verification platform for banking and digital onboarding applications, leveraging Object Detection, Image Segmentation, OCR, and FastAPI to reduce manual verification effort by over **50%**.
- **Engineered automated document-processing pipelines** integrating AWS Textract and PostgreSQL to extract, validate, and structure customer information from identity documents, enabling scalable and reliable data processing.
- **Improved document detection accuracy** through advanced data augmentation, class balancing, and model optimization techniques, significantly reducing false negatives for challenging categories such as ATM cards and mobile devices.
- **Built and optimized 3D deep learning models** for DICOM-based medical image analysis, increasing cancer detection accuracy from **60% to 75%** and improving identification of cancerous regions in volumetric scans.

Projects

ArmyFit AI – Multi-Agent Fitness & Document Intelligence System | Agent Orchestration, Advanced RAG

- **Architected and deployed** an enterprise-grade Multi-Agent AI system using LangGraph, orchestrating Supervisor, RAG, and SQL Agents to automate document intelligence and fitness knowledge workflows.
- **Built an advanced RAG pipeline** using OpenAI, PGVector, semantic retrieval, reranking, and optimized chunking techniques, improving retrieval accuracy and contextual relevance across domain-specific knowledge sources.
- **Developed MCP-enabled SQL Agents** that enabled natural language interaction with PostgreSQL databases, providing seamless access to structured enterprise data.
- **Enhanced production readiness** through Guardrails, short-term and long-term memory, and LangFuse-based monitoring, observability, and evaluation for secure, observable, and reliable AI agent operations.

Vision-Based Quality Inspection for Electronic Circuits | YOLOv11, Pandas, AWS Textract, FastAPI, Kubernetes

- **Architected and deployed** an end-to-end document intelligence platform leveraging YOLOv11, AWS Textract, Pandas, FastAPI to automate extraction, validation, and processing of engineering circuit diagrams from complex PDF documents.
- **Developed a Computer Vision and OCR pipeline** that detected electronic gates, extracted associated metadata, and executed 86 client-defined validation rules to ensure engineering design accuracy and compliance.
- **Improved system performance by 40%** (reducing processing time from 8 to 5 minutes) through model quantization, parallel processing, and optimized data-processing workflows, enabling faster large-scale document analysis.
- **Built a scalable production architecture** using FastAPI, WebSockets, Redis, Celery, Kubernetes, MongoDB, and AWS services, enabling real-time workflow visibility, reliable task execution, and high system availability.

RAG-Based System for Automated PDF Error Analysis | Qdrant, LangChain, FastAPI, Advanced RAG, LightRAG

- **Designed and deployed an advanced RAG platform** using LangChain, OpenAI, and Qdrant Vector Database to automate error diagnosis and knowledge retrieval from large technical document repositories.
- **Implemented advanced retrieval pipelines** including semantic chunking, indexing, reranking, and context-aware response generation, improving answer accuracy and reducing hallucinations.
- **Reduced manual troubleshooting effort** by enabling users to instantly retrieve error-specific solutions through natural language queries instead of manually searching extensive PDF documentation.
- **Enhanced retrieval quality** using LightRAG and Neo4j-based GraphRAG architectures, enabling relationship-aware knowledge discovery and contextual understanding with secure on-premise deployment and low-latency inference.

CrackCell Vision – Electroluminescence Crack Detection System | Computer Vision, YOLO, OpenCV, Django, AWS

- **Designed and deployed** an end-to-end Computer Vision platform using YOLO, OpenCV, Django, SQL Server, and AWS to automate crack detection and quality inspection of solar cells.
- **Improved model performance from 79% to 94% F1-score** through advanced data augmentation, class balancing, and custom training strategies, significantly enhancing defect detection accuracy.
- **Reduced inspection time** from 8–10 minutes to under 20 seconds per image by replacing manual analysis with an AI-powered automated detection pipeline.
- **Developed OpenCV-based image processing workflows** for automated annotation, visualization, and reporting, while collaborating with ASP.NET and frontend teams to integrate the solution into production systems.

Technical Skills

- **Programming:** Python, SQL
- **Generative AI & Agentic AI:** OpenAI, Claude, Gemini, Llama, LangChain, LangGraph, MCP, A2A, LLM Finetuning, PEFT, LORA, Multi-Agent Systems, AI Agents, Tool Calling, Prompt Engineering, Guardrails
- **RAG & Retrieval:** Advanced RAG, GraphRAG, LightRAG, Hybrid Search, Semantic Search, Reranking, Semantic Chunking, Retrieval Optimization
- **Machine Learning & Computer Vision:** PyTorch, TensorFlow, Pandas, NumPy, Scikit-Learn, Hugging Face, YOLO, OpenCV, OCR, AWS Textract, Object Detection, Image Segmentation, MediaPipe
- **Databases & Vector Stores:** PostgreSQL, Redis, Neo4j, PGVector, Qdrant, Pinecone, FAISS, Milvus, ChromaDB
- **Backend & Cloud:** FastAPI, Django, REST APIs, WebSockets, Docker, Kubernetes, AWS, ECS, EKS
- **MLOps & LLMOps:** LangFuse, MLflow, Model Monitoring, Experiment Tracking, Observability, Model Deployment

Certifications

- Building AI Agent & Agentic AI System via Microsoft Autogen (Udemy)
- Generative AI with LLMs (Coursera)
- AWS Networking Masterclass (Udemy)

Education
